BAYESIAN ESTIMATION OF RISK-NEUTRAL DENSITY

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Available at https://github.com/avlsv/Risk-Neutral-Probability

OVERVIEW

This work

- replicates the Bayesian approach to estimating risk-neutral probability density with Dirichlet prior (Fisher, 2016; Härdle, López-Cabrera, and Teng, 2015)
- extends the interpretation of the concentration coefficient as a measure of uncertainty in the risk-neutral probability density.
- applies the method to AAPL stock around the “Liberation Day”.

IMPLIED RISK-NEUTRAL DENSITY

Risk-Neutral Density (Probability) = Implied Probability Density = State-Price Density (SPD) = Risk-Neutral Probability Measure (RNM) = Arrow-Debreu (securities) prices

$$S = \sum_{s=1}^{K} \phi(s) q(s)$$

$$\phi^* \stackrel{d}{=} \frac{\phi(s)}{\sum_{s=1}^{K} \phi(s)} = \frac{\phi(s)}{1/(1+r)} = (1+r)\phi(s)$$

$$S = \frac{1}{1+r} \sum_{s=1}^{K} \phi^*(s) q(s) = \frac{1}{1+r} \mathbb{E}^*[q(s)].$$

IMPLIED RISK-NEUTRAL DENSITY

For TAS utility, the state-prices are

$$\phi(s) = \pi \frac{\beta u'[c(s)]}{u'[c(0)]},$$

$\rightsquigarrow$ implied risk-neutral probability is

$$\phi^*(s) = \pi \frac{\beta u'[c(s)]}{u'[c(0)]}(1+r) = \pi \frac{u'[c(s)]}{u'[c(0)]} \approx \pi,$$

for u' is constant (risk-neutrality) or $c(s) \approx c(0)\ \forall s$.

BREEDEN-LITZENBERGER (1978) FORMULA

For short-term, $r \approx 0$.

$$\begin{aligned} C(S,t) &= \mathbb{E}^* \left[[S(T) - \mathcal{K}]^+ \mid S(t) = S \right] \\ &= \int_0^{+\infty} (x - \mathcal{K})^+ \, dP^*(x) \\ &= \int_{\mathcal{K}}^{+\infty} x \, dP^*(x) - \mathcal{K} \int_0^{+\infty} \, dP^*(x) \\ &= \int_{\mathcal{K}}^{+\infty} x \, dP^*(x) - \mathcal{K}(1 - P^*(\mathcal{K})) \end{aligned}$$

$$\begin{aligned} \frac{\partial C}{\partial \mathcal{K}} &= -\mathcal{K} p^*(\mathcal{K}) - 1 + P^*(\mathcal{K}) + \mathcal{K} p^*(\mathcal{K}) = P^*(\mathcal{K}) - 1 \\ \frac{\partial^2 C}{\partial \mathcal{K}^2} &= p^*(\mathcal{K}). \end{aligned}$$

We need an option price, C, to be twice differentiable, C^2, function.

APPROACHES TO NONPARAMETRIC ESTIMATION

How can we measure the risk-neutral probability density?

- Local polynomial estimation of $C(S,t)$ with shape restrictions – $\frac{\partial^2}{\partial S^2}\hat{C}(S,t)$ is a valid density + Merton (1973) arbitrage restrictions

  (Ait-Sahalia and Duarte, 2003; Lee, 2014)

- Bayesian methods

  (Fisher, 2016; Härdle, López-Cabrera, and Teng, 2015; Taboga, 2016) + KDE in the second case

BAYESIAN FORMULATION

Fisher (2016)[1] proposes the following approach.

$$y_i = \lambda \sum_{j=1}^{K} X_{ij}\beta_j + \varepsilon_i,$$

where $\beta = (\beta_1, \ldots, \beta_K) \in \Delta^{K-1}$, Δ^{K-1} denotes the simplex of dimension $(K-1)$; X_{ij} is a payoff of derivative i in state j, and $\varepsilon_i \sim N(0, \sigma^2)$.

In vector form,

$$y = \lambda X\beta + \varepsilon.$$

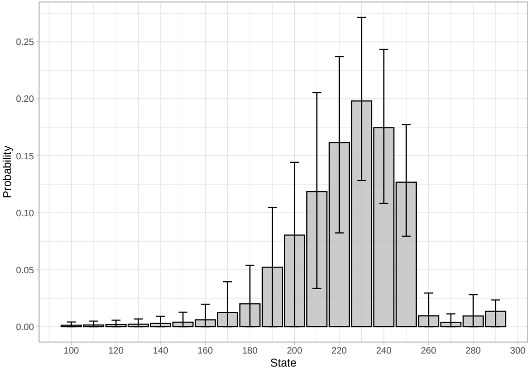
In order to insure $\beta \in \Delta^{K-1}$, we select the (symmetric) Dirichlet prior:

$$p(\beta \mid \alpha) = \mathrm{Dirichlet}(\beta \mid \alpha) \propto \prod_{i=1}^{K} x_i^{\alpha-1}.$$

[1] This is calculated by the Atlanta Fed, *Market Probability Tracker* (2025).

X consists of n payoffs of K derivatives, including the payoffs price (you can think of it as call option with strike equal zero).

The state space is chosen to ensure that the first and last coefficients are near zero, so the list of possibilities is exhaustive.

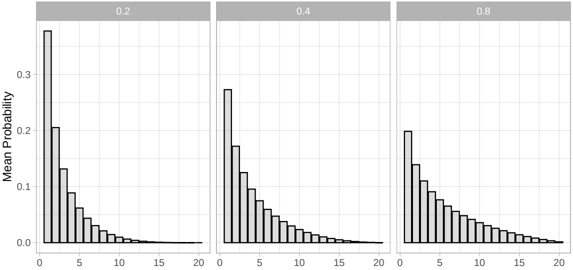


DIRICHLET DISTRIBUTION

The symmetric Dirichlet distribution is a multivariate generalization of the Beta distribution. It is defined on the simplex Δ^{K-1}, which is the set of K-dimensional vectors that sum to 1 and are non-negative.

For $X \sim \mathrm{Dirichlet}(n, \alpha)$

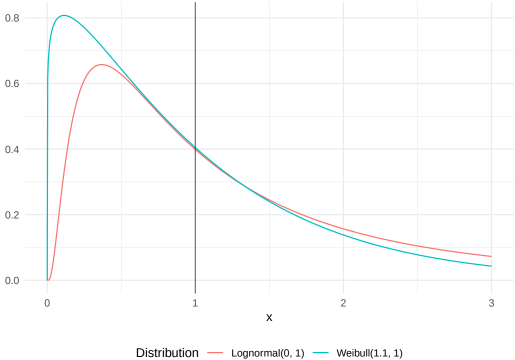
$$\mathbb{E}[X_i] = \frac{1}{n}, \quad \mathrm{var}[X_i] = \frac{n-1}{n}\frac{1}{n(n\alpha+1)}, \quad \mathrm{cov}[X_i, X_{j\neq i}] = \frac{-1}{n(n\alpha+1)}.$$



PRIOR FOR CONCENTRATION COEFFICIENT

α is a vector of concentration parameters for Dirichlet distribution. The lower the value, the less 'flat' the distribution of risk-neutral density β.

It is reasonable to select a distribution that is relatively flat on $[0, 1]$ and assign a large portion of mass to $[0, 1]$. Fisher (2016) uses Lognormal prior, I propose Weibull. The result turns out to be almost identical.



PRIORS AND POSTERIOR

$$p(\beta \mid \alpha) = \text{Dirichlet}(\beta \mid \alpha) \propto \prod_{i=1}^{K} x_i^{\alpha-1}.$$

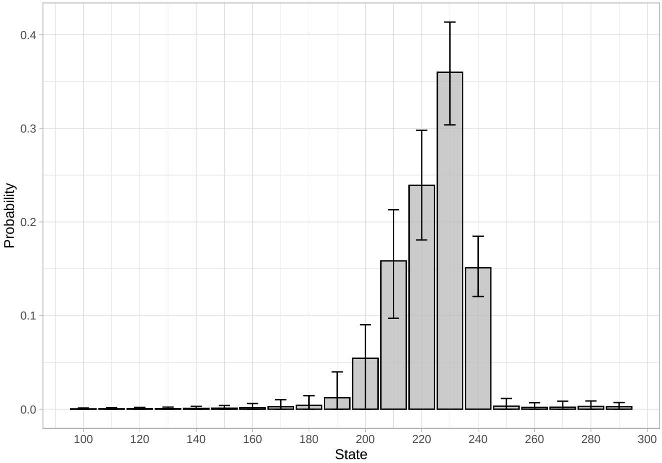
$$p(\alpha) = \text{Weibull}(1.1, 1)$$

$$p(\sigma^2) \propto \frac{1}{\sigma^2}$$

$$p(\alpha, \lambda, \sigma, \beta \mid y) \propto N(y \mid \lambda X\beta, \sigma) \times \text{Dirichlet}(\beta \mid \alpha) \times \text{Weibull}(1.1, 1) \times \frac{1}{\sigma^2}$$

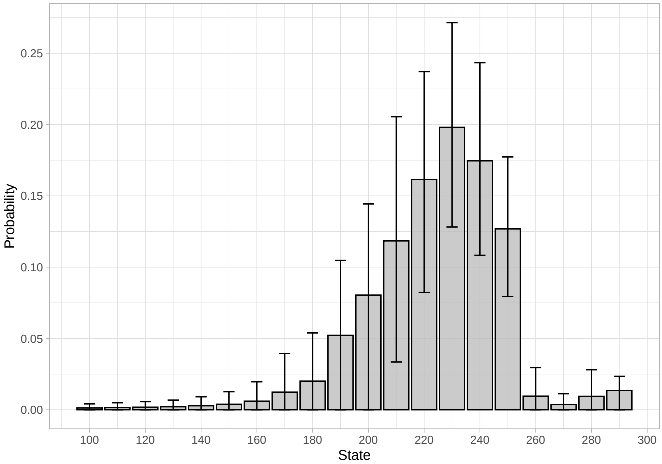
RISK NEUTRAL DENSITY FOR AAPL I

Calculated in 2025-04-01 with expiration date 2025-04-17 (+16 days). Whiskers are 90% HPD interval.



RISK NEUTRAL DENSITY FOR AAPL II

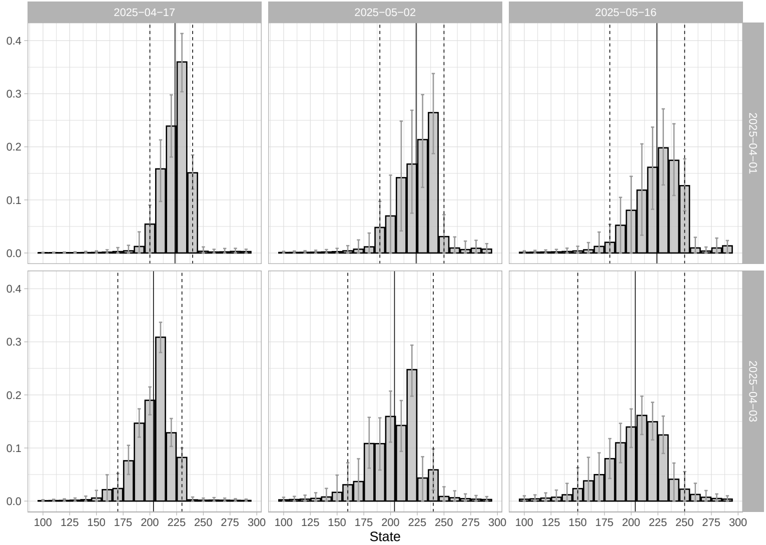
Calculated in 2025-04-01 (same) with expiration date 2025-05-16 (+45 days).



TARIFFS DEBACLE TIMELINE

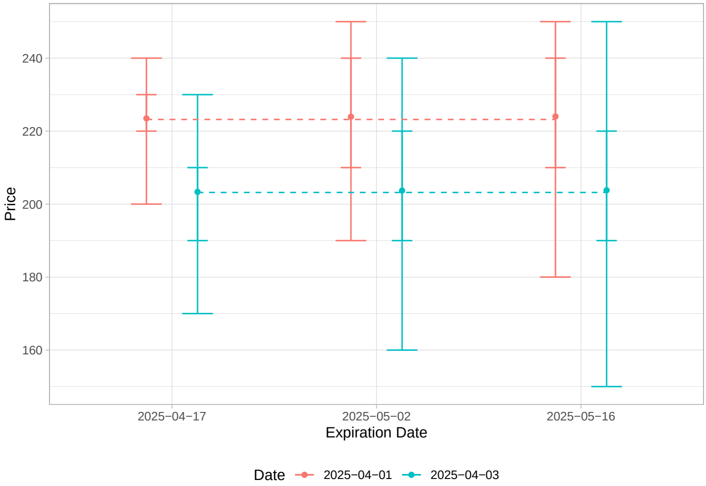
- **April 2.** “Liberation Day”. Tariffs on Chinese goods became 54% (+ 34%)
- **April 4.** China retaliated.
- **April 9.** Trump announced a 90-day pause on universal tariffs.
- **April 11.** Electronics exempted from reciprocal tariffs (tariffs are 20% as before).

DENSITY BEFORE AND AFTER THE "LIBERATION DAY"



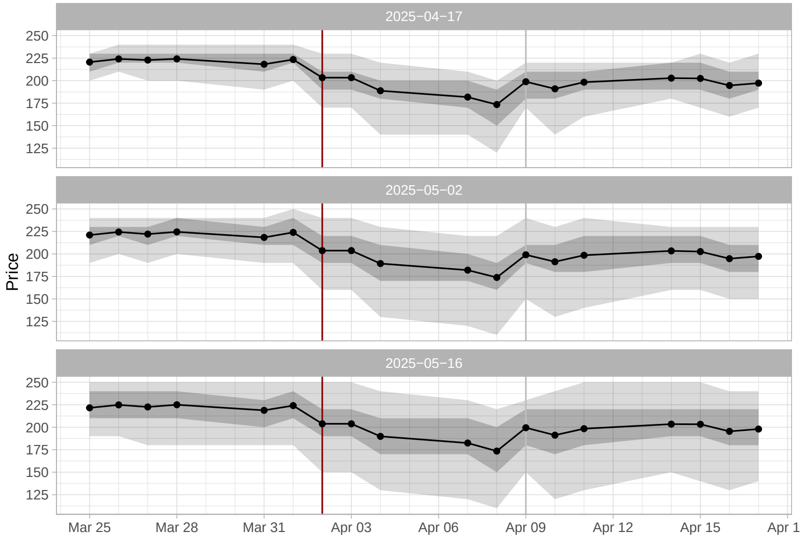
Notes: This figure shows the risk-neutral density of AAPL stock price. The solid line indicates the mean of the distribution, the dashed are 5th and 95th quantiles. The whiskers are 90% HPD intervals.

SUMMARY STATISTICS & SANITY CHECK



- Uncertainty, as indicated by the 5th and 95th percentiles, has increased.
- The price is close to the mean of the distribution.

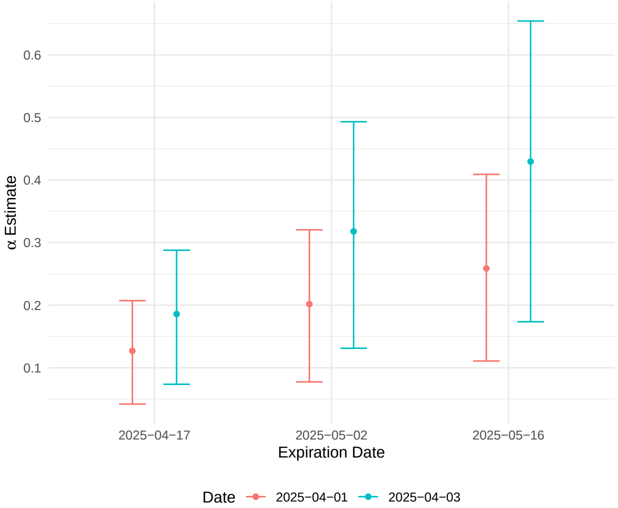
STATE DENSITIES THROUGH TIME



Notes: This figure shows the 5th- 95th percentile interval (in light gray) and 25th-75th percentile interval (in dark gray) of the estimated implied risk neutral density around the liberation date 2025-04-02 (red vertical line). 2025-04-09 is the day of 90 days pause on tariffs (gray vertical line).

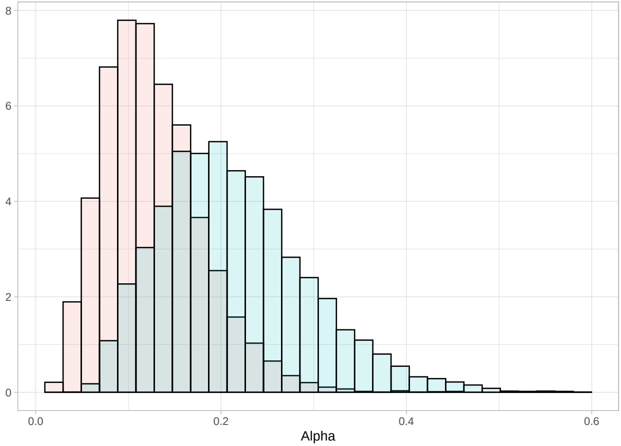
CONCENTRATION PARAMETER POSTERIOR DISTRIBUTION

Concentration parameter, α, is a measure of uncertainty in the risk-neutral probability density. The lower the value, the less 'flat' the distribution $\Longrightarrow$ lower uncertainty.



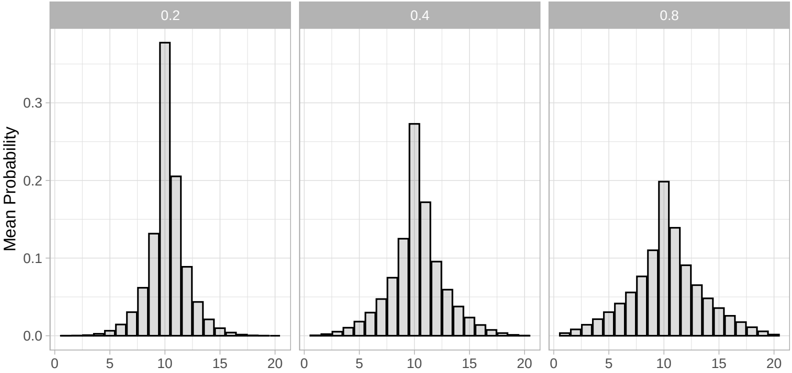
CONCENTRATION PARAMETER POSTERIOR DISTRIBUTION

The posterior distribution of the concentration parameter, α, for expiration 2025-04-17.



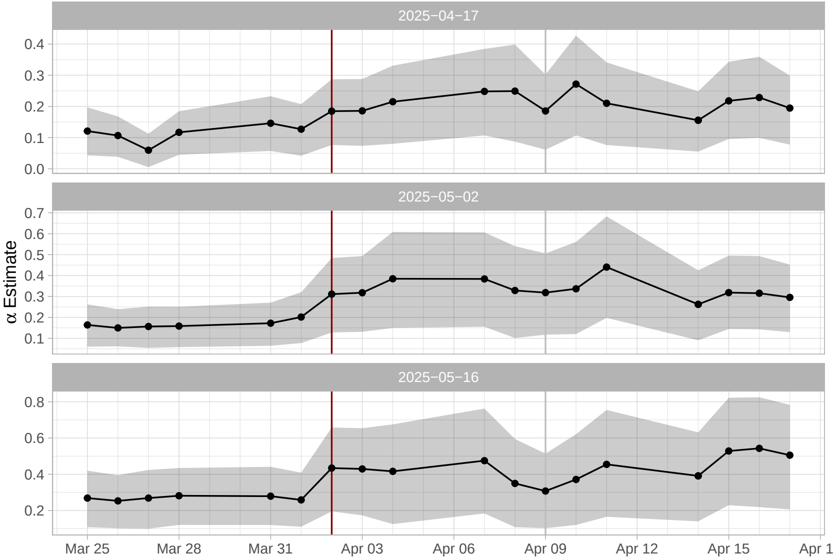
CONCENTRATION PARAMETER AND STATE SPACE

The concentration parameter is dependent on the state space. Since we want to ensure that the list of possibilities is exhaustive, we will be facing several betas close to zero. This corresponds to the concentration parameter being close to zero.



Notes: This figure shows the means of random samples from Dirichlet distribution, that are arranged in order to resemble the distribution with central tendency.

CONCENTRATION PARAMETER POSTERIORS



Notes: This figure shows the posterior mean and HPD interval (in gray) of the concentration parameter of the Dirichlet distribution, α, around the liberation day, 2025-04-02 (red vertical line). 2025-04-09 is the day of 90 days pause on tariffs (gray vertical line).

CONCLUSIONS AND FURTHER WORK

- Concentration parameter is a natural way to measure uncertainty in the risk-neutral probability density.
- The liberation day (April 2) increased the uncertainty of expected AAPL price.

Applications:

- This method can be applied to derivatives of rate of interest (e.g SOFR options by CME/other).
- Concentration parameter then measures whether the forward guidance is effective/was there misinterpretation of signal.

- The concentration parameter α distribution can be conditioned. (What increases uncertainty?)

REFERENCES [1]

Ait-Sahalia, Yacine and Jefferson Duarte (Sept. 2003). "Nonparametric Option Pricing under Shape Restrictions". In: *Journal of Econometrics* 116(1), pp. 9–47. ISSN: 0304-4076. DOI: 10.1016/S0304-4076(03)00102-7. (Visited on 05/31/2025).

Breeden, Douglas T. and Robert H. Litzenberger (1978). "Prices of State-Contingent Claims Implicit in Option Prices". In: *The Journal of Business* 51(4), pp. 621–651. ISSN: 0021-9398. JSTOR: 2352653. (Visited on 05/31/2025).

Fisher, Mark (May 2016). *Simplex Regression*.

Härdle, Wolfgang Karl, Brenda López-Cabrera, and Huei-Wen Teng (Sept. 2015). "State Price Densities Implied from Weather Derivatives". In: *Insurance: Mathematics and Economics* 64, pp. 106–125. ISSN: 0167-6687. DOI: 10.1016/j.insmatheco.2015.05.001. (Visited on 06/07/2025).

Lee, Seung Hwan (Oct. 2014). "Estimation of Risk-Neutral Measures Using Quartic B-spline Cumulative Distribution Functions with Power Tails". In: *Quantitative Finance* 14(10), pp. 1857–1879. ISSN: 1469-7688. DOI: 10.1080/14697688.2012.742202. (Visited on 06/09/2025).

Market Probability Tracker (2025). https://www.atlantafed.org/cenfis/market-probability-tracker. (Visited on 06/09/2025).

Merton, Robert C. (1973). "Theory of Rational Option Pricing". In: *The Bell Journal of Economics and Management Science* 4(1), pp. 141–183. ISSN: 0005-8556. DOI: 10.2307/3003143. JSTOR: 3003143. (Visited on 06/21/2025).

Taboga, Marco (Sept. 2016). "Option-Implied Probability Distributions: How Reliable? How Jagged?" In: *International Review of Economics & Finance* 45, pp. 453–469. ISSN: 1059-0560. DOI: 10.1016/j.iref.2016.07.013. (Visited on 06/02/2025).